diminish. Yet, since it takes light a second and a half to travel the quarter million miles between the moon and the earth, you wouldn't immediately see that the moon had disappeared; for a second and a half, it would seem that the tides were receding from a moon that was still shining high overhead as usual. Thus, according to Newton's approach, gravity can affect us before light—gravity can outrun light—and this, Einstein felt certain, was wrong.[12]

And so, around 1907, Einstein became obsessed with the goal of formulating a new theory of gravity, one that would be at least as accurate as Newton's but would not conflict with the special theory of relativity. This turned out to be a challenge beyond all others. Einstein's formidable intellect had finally met its match. His notebook from this period is filled with half-formulated ideas, near misses in which small errors resulted in long wanderings down spurious paths, and exclamations that he had cracked the problem only to realize shortly afterward that he'd made another mistake. Finally, by 1915, Einstein emerged into the light. Although Einstein did have help at critical junctures, most notably from the mathematician Marcel Grossmann, the discovery of *general relativity* was the rare heroic struggle of a single mind to master the universe. The result is the crowning jewel of pre-quantum physics.

Einstein's journey toward general relativity began with a key question that Newton, rather sheepishly, had sidestepped two centuries earlier. How does gravity exert its influence over immense stretches of space? How does the vastly distant sun affect earth's motion? The sun doesn't touch the earth, so how does it do that? In short, how does gravity get the job done? Although Newton discovered an equation that described the effect of gravity with great accuracy, he fully recognized that he had left unanswered the important question of how gravity actually works. In his *Principia,* Newton wryly wrote, "I leave this problem to the consideration of the reader."[13] As you can see, there is a similarity between this problem and the one Faraday and Maxwell solved in the 1800s, using the idea of a magnetic field, regarding the way a magnet exerts influence on things that it doesn't literally touch. So you might suggest a similar answer: gravity exerts its influence by another field, the gravitational field. And, broadly speaking, this is the right suggestion.